

01/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

PST-3

Time : 180 Mins.

Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power

Chemistry: Chemical Bonding and Molecular Structure, The p-Block Elements (Group-13 & 14), Equilibrium

Botany: Cell: The unit of life, Plant kingdom, Biological classification

Zoology: Animal Kingdom, Structural Organization in Animals, Biomolecules

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720 **Mridul M Anand**
720 3 Year Classroom



720 **Ayush Naugraiya**
720 4 Year Classroom



720 **Arghyadeep Dutta**
720 2 Year Classroom



720 **Aryan Yadav**
720 1 Year Classroom



720 **Palansha Agarwal**
720 2 Year Classroom

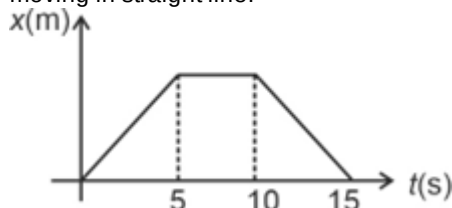


720 **Iram Quazi**
720 1 Year Classroom

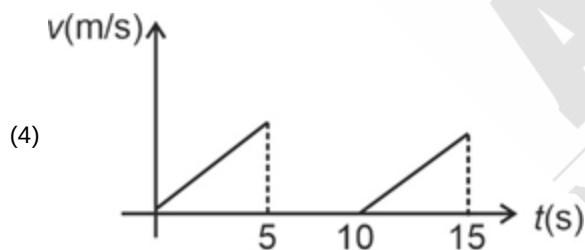
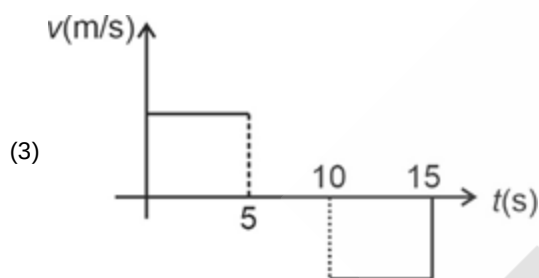
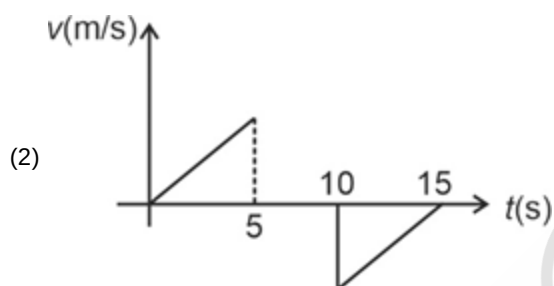
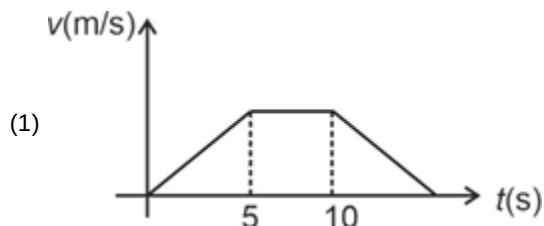
PHYSICS

- Among the following which one is dimensionless error is
 - (1) Systematic error
 - (2) Absolute error
 - (3) Random error
 - (4) Relative error
- If a measured quantity has n significant figure, the reliable digits in it are
 - (1) n
 - (2) $n + 1$
 - (3) $\frac{n}{2}$
 - (4) $n - 1$
- Photon is quantum of a radiation with energy $E = h\nu$ where ν is frequency and h is Planck's constant. The dimension of h is same of that of
 - (1) Linear impulse
 - (2) Angular impulse
 - (3) Force
 - (4) Torque
- Consider the following statement.
Statement A : The dimensional correctness of an equation can be verified using the principle of homogeneity.
Statement B : All unitless quantities are dimensionless.
 Which option is correct based an above information?
 - (1) Both statements A and B are true
 - (2) Both statements A and B are false
 - (3) Statement A is true but statement B is false
 - (4) Statement A is false but statement B is true
- The length of a rectangular plate is measured as 20 cm by Vernier scale of least count 0.02 cm and its breadth as 10 cm by the same scale. The percentage error in area is
 - (1) 0.2%
 - (2) 0.3%
 - (3) 0.5%
 - (4) 0.02%
- Linear momentum (P) associated with a particle is given by $P = (at + bt^2)$, where t is time. The respective dimensions of 'a' and 'b' will be
 - (1) MLT^{-2} and MLT^{-3}
 - (2) M^0LT^{-1} and MLT^{-2}
 - (3) MLT^{-1} and MLT^0
 - (4) MLT^4 and MLT^{-1}
- The power of motor is 100 W. If the unit of length and mass is halved and that of time is also halved, then the power of motor in new system will be
 - (1) 100 W
 - (2) 125 W
 - (3) 50 W
 - (4) 200 W
- Dimensional analysis of relation,
 $(\text{Energy})^n = (\text{Force})^{\frac{1}{2}} (\text{Area})^{\frac{1}{4}}$
 Gives the value of n as
 - (1) $\frac{1}{4}$
 - (2) $\frac{1}{2}$
 - (3) 4
 - (4) 3
- A body is dropped from a height h and it takes t seconds to travel first $\left(\frac{1}{x}\right)^{\text{th}}$ of h . Then total time of descent is
 - (1) $\frac{t}{\sqrt{x}}$
 - (2) $t\sqrt{x}$
 - (3) $\frac{\sqrt{x}}{t}$
 - (4) $\frac{1}{t\sqrt{x}}$

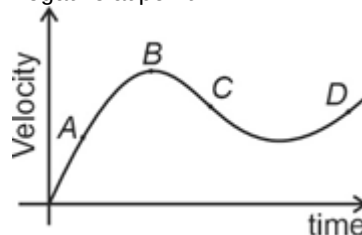
10. Figure shows the position time ($x - t$) graph of a body moving in straight line.



Which one of the graph shown below represents the velocity-time ($v-t$) graph of the motion of the body



11. The velocity-time graph of a moving particle is shown in figure. The instantaneous acceleration of the particle is negative at point



- (1) A
(2) B
(3) C
(4) D
12. A body falls freely from height ' h ', its average velocity when it reaches at $\frac{h}{2}$ is

(1) $\frac{\sqrt{gh}}{2}$

(2) $\sqrt{\frac{gh}{2}}$

(3) $\sqrt{2gh}$

(4) $\sqrt{gh}$

13. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

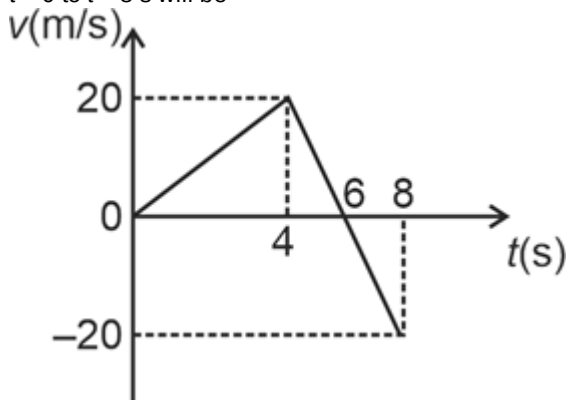
Assertion (A): If the body is slowing down then the angle between velocity and acceleration may be 180° .

Reason (R): If the body is slowing down, the distance covered is negative.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true, (R) is false
(4) (A) is false, (R) is true

14. The velocity-time graph of a body moving in a straight line is shown in figure. The average speed of the body between $t = 0$ to $t = 8$ s will be



- (1) 4 m/s
(2) 5 m/s
(3) 10 m/s
(4) 8 m/s
15. The velocity of a particle is given by, $v = (2t + 3t^2)$ m/s where t is time in second, then distance travelled by the particle between $t = 2$ s to $t = 3$ s will be
- (1) 24 m
(2) 12 m
(3) 36 m
(4) 18 m
16. Two train A and B having length 200 m and x are moving on parallel track in opposite direction with velocity 10 m/s and 15 m/s, if time taken by the trains to cross each other is 10 s, then value of x in metres will be
- (1) 100
(2) 200
(3) 50
(4) 75
17. When an object is dropped from an aeroplane moving horizontally with constant velocity then
- (a) Path of the object relative to earth is parabola
(b) Path of the object relative to pilot is straight line vertically down
(c) Speed of the particle increases continuously
- (1) Only (a) and (b) are correct
(2) Only (b) and (c) are correct
(3) All (a), (b) and (c) are correct
(4) Only (a) and (c) are correct

18. A body is projected at an angle of 60° with the horizontal with momentum P . At its highest point the magnitude of the momentum is

- (1) P
(2) $\frac{\sqrt{3}P}{2}$
(3) $\frac{P}{2}$
(4) $\frac{P}{4}$

19. $\vec{A}$ and $\vec{B}$ are two perpendicular vector. If $|\vec{B}| = 2|\vec{A}|$ then angle between $\vec{B}$ and resultant vectors $\vec{R}$ will be

- (1) $\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$
(2) $\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$
(3) 45°
(4) 37°

20. The wind is blowing from south towards north at 5 m/s but to a cyclist it appears blowing from east to west at 5 m/s. The velocity of the cyclist is

- (1) $5\sqrt{2}$ m/s toward N-E
(2) $5\sqrt{2}$ m/s toward S-W
(3) $5\sqrt{2}$ m/s toward S-E
(4) $5\sqrt{2}$ m/s toward N-W

21. A body is projected with a velocity 80 m/s at an angle 60° with the vertical, the velocity vector of the particle at $t = 1$ s will be

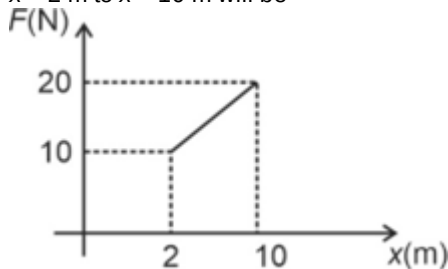
- (1) $(40\sqrt{3}\hat{i} + 30\hat{j})$ m/s
(2) $(20\sqrt{3}\hat{i} + 20\hat{j})$ m/s
(3) $(30\sqrt{3}\hat{i} - 30\hat{j})$ m/s
(4) $(20\sqrt{3}\hat{i} - 10\hat{j})$ m/s

22. Two particles A and B are projected with same speed at angles θ and $90 - \theta$ respectively from horizontal. The ratio of maximum height of A to the maximum height of B will be

- (1) $\cot^2\theta$
(2) $\tan^2\theta$
(3) $\sin^2\theta$
(4) 1

23. A particle starts from origin at $t = 0$ with velocity $(4\hat{i} \text{ m/s})$ and moves in x - y plane with constant acceleration of $(6\hat{i} + 8\hat{j}) \text{ m/s}^2$. The x -coordinate of particle in 3 s will be equal to
- 13 m
 - 17 m
 - 34 m
 - 39 m
24. Consider the following statements.
 (A) In uniform circular motion, the dot product of velocity vector and acceleration vector would be zero.
 (B) In uniform circular motion, the change in magnitude of velocity is zero for a half cycle.
 (C) In uniform circular motion, the magnitude of change in velocity is zero for a half cycle.
 Based on above information, pick right option.
- Only statement (A) is incorrect
 - Only statement (B) is incorrect
 - Only statement (C) is incorrect
 - All statements are incorrect
25. A man can row a boat in still water with velocity 6 km/h and water is flowing in a river with velocity 4 km/h. If width of the river is 1 km, then minimum time taken to cross the river in minutes will be
- 10
 - 5
 - 15
 - 20
26. A body of mass m tied at the end of a string of length l is rotated in a vertical circle. If the velocity of the body at lowest point is $\sqrt{5gl}$, then difference of maximum to minimum tension in the string will be
- $3mg$
 - $6mg$
 - $2mg$
 - $4mg$
27. A body of mass 4 kg moves according to relation $x = (t + 2t^2) \text{ m}$ where t is time in second. The power of the body at $t = 1 \text{ s}$ will be
- 40 W
 - 20 W
 - 80 W
 - 60 W
28. The collision in which the relative velocity of approach is equal to relative velocity of separation is known as
- Perfectly elastic collision
 - Perfectly inelastic collision
 - Inelastic collision
 - May be elastic or inelastic depends on mass of the body
29. Force acting on a particle is given by $(2\hat{i} + 4\hat{j}) \text{ N}$. Work done by this is zero, when the particle moves along the line $4y + bx = 2$, then the value of b will be
- 1
 - 3
 - $-\frac{3}{2}$
 - 2
30. Two springs A and B have their force constants k_A and k_B ($k_A > k_B$). When they are stretched by the same force, work done is
- Same in both springs
 - More in spring A
 - More in spring B
 - None of these
31. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion (A): The work done by conservative force during a round trip is zero.
Reason (R) : No force is required to move a body in its round trip.
 In the light of the above statements, choose the **correct** answer from the options given below:
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (A) is true, (R) is false
 - (A) is false, (R) is true

32. The force displacement graph is shown in the figure. The work done by the force while displacing the object from $x = 2$ m to $x = 10$ m will be



- (1) 40 J
(2) 120 J
(3) 60 J
(4) 80 J
33. When a mass ' M ' is attached to a vertical spring and slowly released, the extension in spring is ' x ' when a mass ' $3M$ ' is attached with the same spring and released suddenly, then extension in the spring will be

- (1) $4x$
(2) $8x$
(3) $3x$
(4) $6x$

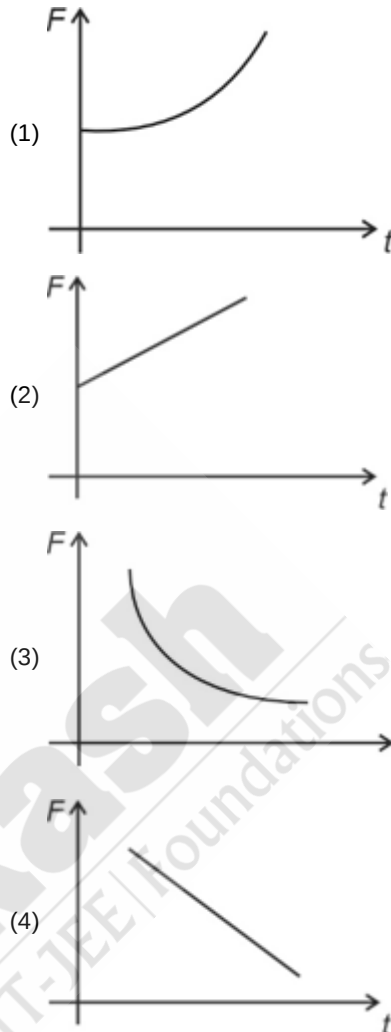
34. A particle of mass m is attached with a vertical string of length l and is given a velocity at its lower most position such that it just complete the circle. On basis of given information match the column I with column II and choose the correct option

	Column I		Column II
a	Radial acceleration at lower most position (in m/s^2)	(i)	g
b	Tension in string at horizontal position	(ii)	Zero
c	Radial acceleration at upper most position (in m/s^2)	(iii)	$3mg$
d	Tension in string at uppermost position (in N)	(iv)	$5g$

- (1) a(ii), b(i), c(iv), d(iii)
(2) a(iv), b(iii), c(i), d(ii)
(3) a(i), b(iv), c(iii), d(ii)
(4) a(i), b(iii), c(ii), d(iv)
35. The magnitude of two vectors are 3 and 4 units and their scalar product is equal to $6\sqrt{3}$ units. The angle between the vectors is

- (1) $\frac{\pi}{3}$ rad
(2) $\frac{\pi}{4}$ rad
(3) $\frac{\pi}{6}$ rad
(4) $\frac{\pi}{2}$ rad

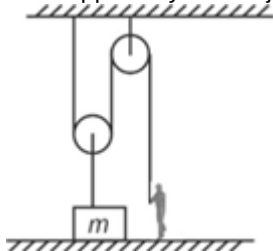
36. The linear momentum of a particle varies with time t as $p = (a_0 + at + bt^3)$. Which of the following graph represents correct force time ($F-t$) relation? (Where a_0 and b are constants)



37. A bomb moving with velocity $(30\hat{i} + 36\hat{j} + 24\hat{k})$ m/s explodes into pieces of mass ratio 1 : 3. If small piece gases out with velocity $(100\hat{i} + 40\hat{j} - 10\hat{k})$ m/s, then velocity of larger piece after explosion will be

- (1) $(\frac{20}{3}\hat{i} + \frac{104}{3}\hat{j} + \frac{106}{3}\hat{k})$ m/s
(2) $(-\frac{10}{3}\hat{i} - \frac{100}{3}\hat{j} + \frac{106}{3}\hat{k})$ m/s
(3) $(-10\hat{i} - 30\hat{j} + 40\hat{k})$ m/s
(4) $(-20\hat{i} - 60\hat{j} - 40\hat{k})$ m/s

38. A man pulls the string as shown in figure, then the minimum force applied by man to just lift the block will be



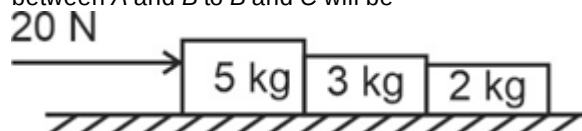
- (1) mg
 (2) $\frac{mg}{4}$
 (3) $\frac{mg}{2}$
 (4) $2mg$
39. A block of mass 4 kg is placed on rough horizontal plane. A time dependent horizontal force $F = (kt)$ N acts on the block, ($k = 4$ N/s). If coefficient of friction between the surface and block is 0.4, then time at which the block will be on the verge of moving is ($g = 10 \text{ m/s}^2$)



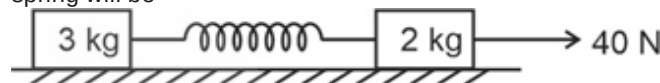
- (1) 2 s
 (2) 1 s
 (3) 3 s
 (4) 4 s
40. The apparent weight of a freely falling body is
- (1) More than the actual weight
 (2) Less than the actual weight
 (3) Equal to actual weight
 (4) Zero
41. Consider the following statements.
 A. Frictional force always opposes motion.
 B. Coefficient of static friction is greater than coefficient of kinetic friction for given pair of surface.
 C. Frictional force is non conservative force.
 Based on above information pick the correct statement.

- (1) Only B is true
 (2) Only B and C are true
 (3) Only C is true
 (4) All statements are true

42. Three blocks A, B and C of mass 5 kg, 3 kg and 2 kg respectively are kept on smooth horizontal surface. If a force of 20 N is applied on 5 kg block, then ratio of contact force between A and B to B and C will be

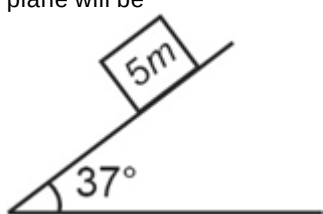


- (1) $\frac{3}{2}$
 (2) 2
 (3) $\frac{5}{2}$
 (4) 3
43. Two blocks of mass 3 kg and 2 kg are connected by a spring having spring constant 400 N/m. The blocks are placed on smooth horizontal surface as shown in figure. If a force of 40 N applied on 2 kg block, then extension in the spring will be

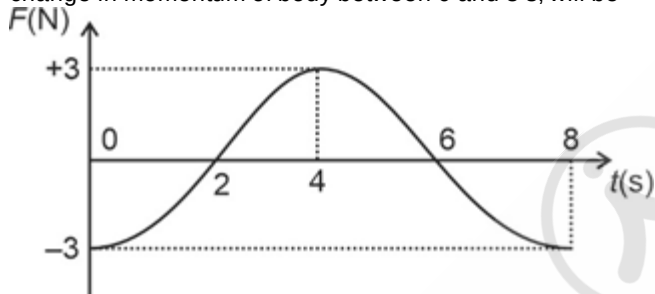


- (1) 4 cm
 (2) 2 cm
 (3) 8 cm
 (4) 6 cm

44. A block of mass 5 kg is placed on rough inclined plane of angle of inclination 37° is just on the verge of moving, then the coefficient of friction between the block and inclined plane will be



- (1) $\frac{3}{5}$
 (2) $\frac{4}{5}$
 (3) $\frac{3}{4}$
 (4) $\frac{1}{4}$
45. A force time graph for the motion of a body is shown. The change in momentum of body between 0 and 8 s, will be



- (1) 6 N-s
 (2) 12 N-s
 (3) Zero
 (4) None of these

CHEMISTRY

46. Match List-I with List-II

	List-I Species		List-II Dipole moment, $\mu(D)$
a.	H ₂ O	(i)	0.95
b.	H ₂ S	(ii)	0.23
c.	NH ₃	(iii)	1.85
d.	NF ₃	(iv)	1.47

The correct match is

- (1) a(iv), b(iii), c(i), d(ii)
 (2) a(iii), b(i), c(iv), d(ii)
 (3) a(iii), b(iv), c(i), d(ii)
 (4) a(iv), b(i), c(iii), d(ii)

47. Given below are two statements

Statement I : Bond enthalpy of N₂ molecule is higher than O₂ molecule.

Statement II : Bond enthalpy of HCl molecule is higher than H₂ molecule.

In the light of above statements, choose the correct answer from the options given below

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

48. For the reaction, $\text{CaCO}_3(s) \rightleftharpoons \text{CaO}(s) + \text{CO}_2(g)$ $K_p = 1.64$ atm at 527°C . If 100 g of CaCO_3 were taken at 527°C in a 5L vessel, then amount of CO_2 formed at equilibrium will be
- 5.5 g
 - 1.2 g
 - 8.7 g
 - 4.2 g
49. Consider the following statements
- Ti^+ is more stable in solution than Ti^{3+} ion
 - Aluminium reacts with aqueous NaOH to form sodium tetrahydroxoaluminate (III) and hydrogen gas
 - Aluminium chloride in acidified aqueous solution forms tetrahedral $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$ ion only
- Choose the correct statement(s)
- (b) and (c) only
 - (a) and (c) only
 - (a) and (b) only
 - (c) only
50. Which of the following reactions does not represent the formation of boric acid?
- $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow$
 - $\text{BCl}_3 + \text{H}_2\text{O} \rightarrow$
 - $\text{Na}_2\text{B}_4\text{O}_7 + \text{H}_2\text{O} \rightarrow$
 - $\text{B}_2\text{H}_6 + \text{O}_2 \xrightarrow{\Delta}$
51. Match List-I with List-II
- | | List-I
Species | | List-II
Bond order |
|----|-------------------|-------|-----------------------|
| a. | N_2 | (i) | 1.0 |
| b. | CO | (ii) | 0.5 |
| c. | O_2^{2-} | (iii) | 2.5 |
| d. | F_2 | (iv) | 3.0 |
- The correct match is
- a(ii), b(iii), c(i), d(iv)
 - a(iii), b(iv), c(i), d(ii)
 - a(i), b(iii), c(ii), d(iv)
 - a(iv), b(ii), c(i), d(iii)
52. Consider the following statements
- Both XeF_4 and SF_4 have see-saw shape
 - Highest occupied molecular orbital (HOMO) of N_2 molecule has one node
 - In ethyne molecule, each carbon atom is sp hybridised
 - Both BrF_5 and ClF_3 have two lone pairs on central atom each
- The incorrect statement(s) is/are
- (a), (b) and (d) only
 - (a) and (c) only
 - (b) and (c) only
 - (d) only
53. Consider the following molecules PF_5 , BF_3 , IF_7 , NO_2 , H_3PO_4 , SF_4 , SO_3
- The total number of molecules which have expanded octet is
- 5
 - 3
 - 4
 - 6
54. For which of the following options, an increase in volume will move the reaction in backward direction
- $2\text{NH}_3(g) \rightleftharpoons \text{N}_2(g) + 3\text{H}_2(g)$
 - $\text{N}_2\text{O}_4(g) \rightleftharpoons 2\text{NO}_2(g)$
 - $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$
 - $\text{CO}_2(g) + \text{C}(s) \rightleftharpoons 2\text{CO}(g)$
55. If K_a for a monobasic acid is 10^{-5} , then value of $\text{p}K_b$ for its conjugate base at 298 K is
- 5
 - 7
 - 8
 - 9
56. Given below are two statements
- Statement I :** The solubility product of a saturated solution of $\text{Cd}(\text{OH})_2$ is 2.5×10^{-14} if its pOH is 4.
- Statement II :** $\text{Ni}(\text{OH})_2$ is more soluble in water than NiS (K_{sp} of $\text{Ni}(\text{OH})_2 = 2.0 \times 10^{-15}$, K_{sp} of $\text{NiS} = 4.7 \times 10^{-5}$)
- In the light of above statements, choose the correct answer from the options given below
- Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
 - Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct

57. Match List-I with List-II

	List-I Species		List-II Nature
a.	CO	(i)	Basic
b.	GeO ₂	(ii)	Amphoteric
c.	Tl ₂ O ₃	(iii)	Neutral
d.	Ga ₂ O ₃	(iv)	Acidic

The correct match is

- (1) a(ii), b(iii), c(iv), d(i)
 (2) a(iii), b(ii), c(i), d(iv)
 (3) a(ii), b(i), c(iii), d(iv)
 (4) a(iii), b(iv), c(i), d(ii)

58. Given below are two statements

Statement I : SiCl₄ undergoes hydrolysis to form silicic acid.

Statement II : The correct order of catenation of group 14 elements is C >> Si > Ge ≈ Sn

In the light of above statements, choose the correct answer from the options given below

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

59. The correct decreasing order of bond enthalpy is

- (1) Sn – Sn > Ge – Ge > Si – Si > C – C
 (2) Si – Si > Ge – Ge > C – C > Sn – Sn
 (3) C – C > Si – Si > Ge – Ge > Sn – Sn
 (4) Ge – Ge > Si – Si > C – C > Sn – Sn

60. In which of the following processes, both magnetic behaviour as well as bond order changes?

- (1) O₂ → O₂⁺
 (2) C₂ → C₂⁺
 (3) B₂ → B₂⁺
 (4) O₂ → O₂⁻

61. Consider the following statements

- (a) Both CO₃²⁻ and CO₂ have three canonical forms
 (b) Formal charge on central oxygen atom in ozone is zero
 (c) Formal charge on nitrogen in HNO₃ is +1
 (d) ZnCl₂ has more covalent character than KCl
 Choose the correct statement(s)

- (1) (a), (c) and (d) only
 (2) (a) only
 (3) (b) and (d) only
 (4) (b) and (c) only

62. The correct order of covalent bond length is

- (1) C – H > C – O > O – H > N – O
 (2) C – O > N – O > C – H > O – H
 (3) C – H > O – H > N – O > C – O
 (4) O – H > C – H > C – O > N – O

 63. The bond order of N – O bond and P – O bond in NO₃⁻ and PO₄³⁻ ion respectively are

- (1) 1.33 and 1.25
 (2) 1.5 and 1.25
 (3) 1.25 and 1.5
 (4) 1.5 and 1.33

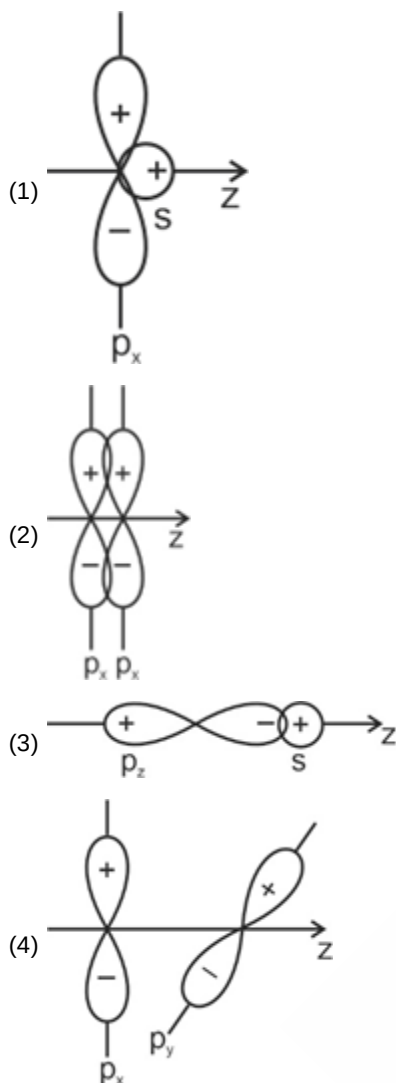
64. Match List-I with List-II

	List-I Species		List-II Hybridisation and bond angle
a.	H ₂ O	(i)	sp ² and 120°
b.	SO ₂	(ii)	sp ³ and 104.5°
c.	NH ₃	(iii)	sp ² and 119.5°
d.	BF ₃	(iv)	sp ³ and 107°

The correct match is

- (1) a(iv), b(iii), c(ii), d(i)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(iv), b(i), c(ii), d(iii)
 (4) a(ii), b(iii), c(iv), d(i)

65. Which of the following represents negative overlapping?



66. Choose the incorrect statement about PCl_5 molecule.

- (1) It has sp^3d hybridisation
- (2) Equatorial $\text{P}-\text{Cl}$ bonds slightly longer than axial $\text{P}-\text{Cl}$ bonds
- (3) It has three ($\text{Cl}-\text{P}-\text{Cl}$) 120° bond angles and six ($\text{Cl}-\text{P}-\text{Cl}$) 90° bond angles
- (4) It has trigonal bipyramidal geometry

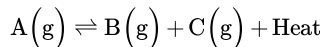
67. 100 mL of 0.01 M solution of NaOH is mixed with 100 mL of 0.01 M solution of H_2SO_4 . The pH of resultant solution will be

- (1) 4.1
- (2) 2.9
- (3) 3.2
- (4) 2.3

68. pH of 0.02 M sodium acetate solution is (pK_a of CH_3COOH = 4.74)

- (1) 7.51
- (2) 10.87
- (3) 8.52
- (4) 9.86

69. For the reversible reaction,



The equilibrium shifts in the backward direction by

- (1) Increasing the concentration of A(g)
- (2) Increasing the temperature
- (3) Increasing the volume of closed vessel
- (4) Decreasing the concentration of C(g)

70. Consider the following statements

- (a) In borax, all boron atoms are sp^3 hybridised.
- (b) Boric acid is weak monobasic Lewis acid.
- (c) Boric acid has a layer structure in which planar BO_3 units are joined by hydrogen bonds.
- (d) Borax is heated in a Bunsen burner flame with CoO on a loop of platinum wire, a blue coloured $\text{Co(BO}_2)_2$ bead is formed.

The correct statement(s) is/are

- (1) (a) and (c) only
- (2) (b) only
- (3) (a), (b) and (c) only
- (4) (b), (c) and (d) only

71. The percent ionisation of 0.02 M acetic acid in 0.2 M HCl will be nearly (K_a of acetic acid is 1.74×10^{-5})

- (1) 0.009%
- (2) 0.08%
- (3) 0.002%
- (4) 0.02%

72. Consider the following statements

- (a) Buckminsterfullerene contains twelve six-membered rings and twenty five-membered rings
- (b) Graphite is thermodynamically more stable than diamond
- (c) Charcoal is obtained by heating wood at high temperatures in the absence of air

The correct statement(s) is/are

- (1) (a) and (b) only
- (2) (a) only
- (3) (b) and (c) only
- (4) (a) and (c) only

73. Given below are two statements

Statement I: Hydrolysis of dimethyldichlorosilane followed by condensation polymerisation yields straight chain polymers.

Statement II : On heating, orthoboric acid above 370K forms metaboric acid, HBO_2 which on further heating yields boric oxide, B_2O_3 .

In the light of above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

74. If the equilibrium constant value of a reaction is 200 at 27°C , then the standard Gibbs energy change of the reaction will be nearly

- (1) -11.49 kJ/mole
- (2) -13.22 kJ/mole
- (3) -12.87 kJ/mole
- (4) -21.42 kJ/mole

75. Consider the following statements

- (a) ZSM-5 is used to convert alcohols directly into gasoline
- (b) The basic structural unit of silicates is SiO_4^{4-}
- (c) Hydrated zeolites are used as ion exchangers in softening of "hard water"
- (d) Silicon dioxide is a covalent, three-dimensional network solid

The correct statements are

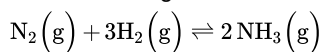
- (1) (a) and (d) only
- (2) (b), (c) and (d) only
- (3) (a), (b) and (c) only
- (4) (a), (b), (c) and (d)

76. Find out K_a for 10^{-1} M HCN having pOH equal to 10?

- (1) 10^{-3}
- (2) 10^{-4}
- (3) 10^{-6}
- (4) 10^{-7}

77. Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A) : Addition of an inert gas at constant volume to the following reaction at equilibrium



Favours the reaction in forward direction

Reason (R) : Addition of an inert gas at constant volume shifts equilibrium towards lesser number of gaseous moles
In the light of above statements, choose the most appropriate answer from the options given below

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

78. pH of a solution obtained by mixing 200 mL of 0.1 M CH_3COOH with 100 mL of 0.05 M NaOH will be (pK_a for $\text{CH}_3\text{COOH} = 4.74$)

- (1) 4.26
- (2) 4.74
- (3) 5.27
- (4) 3.74

79. Consider the following statements about carbon monoxide

- (a) The complex of carbon monoxide with haemoglobin is about 300 times more stable than the oxygen-haemoglobin complex
 - (b) On small scale, pure carbon monoxide is prepared by dehydration of formic acid with concentrated H_2SO_4 at 373 K
 - (c) It is a blue colour, odourless and highly water- soluble gas
- The correct statement(s) is/are

- (1) (a), (b) and (c)
- (2) (b) only
- (3) (b) and (c) only
- (4) (a) and (b) only

80. The ionization constant of ammonium hydroxide and acetic acid are 1.77×10^{-5} and 1.8×10^{-5} at 298 K respectively. Hydrolysis constant of ammonium acetate is

- (1) 1.6×10^{-4}
- (2) 2.5×10^{-4}
- (3) 3.1×10^{-5}
- (4) 1.9×10^{-4}

81. Given below are two statements

Statement I : In diborane, maximum number of atoms present in a plane is six

Statement II : In diborane, normal 2-centre-2- electron bonds and banana bonds are two and four respectively
In the light of above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

82. Choose the incorrect match

	Conjugate acid	Conjugate base
(1)	HCO_3^-	CO_3^{2-}
(2)	H_2PO_2^-	HPO_2^-
(3)	H_2PO_4^-	HPO_4^{2-}
(4)	HPO_4^{2-}	PO_4^{3-}

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

83. The pair of compounds which can make basic buffer is

- (1) 100 mL of 0.1 M NH_4OH and 200 mL of 0.1 M HCl
- (2) 100 mL of 0.2 M NH_4OH and 50 mL of 0.2 M HCl
- (3) 200 mL of 0.1 M HCOOH and 100 mL of 0.2 M NaOH
- (4) 200 mL of 0.2 M HCOOH and 200 mL of 0.1 M NaOH

84. Given below are two statements

Statement I : If $Q_C > K_C$, the reaction will proceed in the backward direction.

Statement II : If $Q_C < K_C$, the reaction will proceed in the forward direction

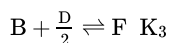
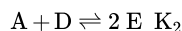
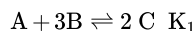
In the light of above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

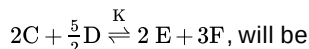
85. The solubility of AgCl in the presence of 0.1 M NaCl solution is ($K_{sp}(\text{AgCl}) = 1.8 \times 10^{-10}$)

- (1) 4.2×10^{-8} M
- (2) 2.6×10^{-7} M
- (3) 1.8×10^{-9} M
- (4) 1.2×10^{-8} M

86. The equilibrium constants of the following are



The equilibrium constant (K) of the reaction



- (1) $K_1 K_2 K_3^2$
- (2) $\frac{K_2^2 K_3}{K_1}$
- (3) $\frac{K_2^3 K_3}{K_1}$
- (4) $\frac{K_2 K_3^3}{K_1}$

87. Match List-I with List-II

	List-I Order		List-II Property
a.	$\text{B} > \text{Al} > \text{Tl} > \text{In} > \text{Ga}$	(i)	Boiling point
b.	$\text{B} > \text{Al} > \text{Ga} > \text{In} > \text{Tl}$	(ii)	Electronegativity
c.	$\text{B} > \text{Tl} > \text{Ga} > \text{Al} > \text{In}$	(iii)	Melting point
d.	$\text{B} > \text{Tl} > \text{In} > \text{Ga} > \text{Al}$	(iv)	First ionization enthalpy

The correct match is

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(iii), b(i), c(iv), d(ii)
- (3) a(iv), b(i), c(ii), d(iii)
- (4) a(ii), b(iii), c(iv), d(i)

88. Given below are two statements

Statement I : According to molecular orbital theory, both C_2 and B_2 species have only π bonds

Statement II : Both $\pi 2p_x$ and $\pi 2p_y$ molecular orbitals of B_2 and C_2 have 2 electrons each

In the light of above statements, choose the correct answer from the options given below

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

89. If $K_w = 10^{-12}$ at 90°C , then the pH of pure water at 90°C will be
- (1) 6
 - (2) 12
 - (3) 8
 - (4) 7
90. Consider the following statements
Statement I : Both H_2O and NH_3 molecules have intermolecular hydrogen bonding
Statement II : Both o-nitrophenol and salicylic acid have intramolecular hydrogen bonding
 In the light of above statements, choose the correct answer from the options given below.
- (1) Both statement I and statement II are correct
 - (2) Both statement I and statement II are incorrect
 - (3) Statement I is correct but statement II is incorrect
 - (4) Statement I is incorrect but statement II is correct

BOTANY

91. Who first explained "*Omnis cellula-e cellula*"?
- (1) Schleiden
 - (2) Schwann
 - (3) Rudolf Virchow
 - (4) Robert Brown
92. In bacteria, mesosomes are present in all of the following forms, except
- (1) Vesicles
 - (2) Tubules
 - (3) Cisternae
 - (4) Lamellae
93. Which of the following statements is **incorrect** w.r.t. the outermost layer of prokaryotic cell envelope?
- (1) The layer differs in thickness and chemical composition in different bacteria
 - (2) It protects the cell from loss of water and nutrients
 - (3) It protects the host's immune system from adverse effects of bacteria
 - (4) It protects the bacterial cell and also helps in adhesion
94. Find the **odd** one w.r.t. bacterial flagellum.
- (1) It is a thin filamentous extension from the cell wall
 - (2) It is composed of three parts – filament, hook and basal body
 - (3) Filament is the longest portion
 - (4) It is a hollow, rigid, cylindrical structure made up of protein filaments called tubulin
95. Which of the following structures are involved in the mating process in Gram-negative bacteria?
- (1) Fimbriae
 - (2) Pili
 - (3) Polysomes
 - (4) Mesosomes

96. Identify the following statements as true (T) or false (F) and choose the **correct** option.?
- Partially embedded integral membrane proteins are called trans membrane proteins.
 - Secondary wall in plants is a multi-layered structure, which gradually diminishes with maturity.
 - Middle lamella between adjacent cells in plant tissue is a thin, sticky and amorphous layer of cementing material.

	a	b	c
(1)	T	F	T
(2)	T	T	F
(3)	F	F	T
(4)	F	T	F

- (1)
 - (2)
 - (3)
 - (4)
97. Proponents of artificial system of classification of organisms were
- Aristotle and Linnaeus
 - George Bentham and Dalton Hooker
 - Engler and Prantl
 - Takhtajan and Hutchinson
98. The branch of taxonomy, which is based on the chemical constituents of the plant, uses _____ to resolve confusion in classification.
Choose the **correct** option to fill in the blank
- Phenetics
 - Chromosome structure and behaviour
 - Natural affinities amongst the organisms
 - DNA sequence
99. Anisogamous mode of sexual reproduction occurs in
- Ulothrix*
 - Eudorina*
 - Volvox*
 - Polysiphonia*
100. Which of the following is **incorrectly** matched?
- Colonial algae - *Volvox*
 - Pyrenoids - Chloroplast
 - Two equal anteriorly attached flagella in zoospores - Phaeophyceae
 - Kelp - *Laminaria*

101. Which of the following features are associated with algae showing predominance of r-phycoerythrin?
- Floridean starch as stored food
 - Oogamous sexual reproduction
 - Coating of algin
 - Complex post-fertilization developments
 - Polysulphate esters in the cell wall
- Choose the **correct** answer from the following options.

- b, c and d only
 - a, b, d and e only
 - a and e only
 - b, d and e only
102. All of the following are the exclusive features of the members of the kingdom Monera, **except**
- Presence of 70S cytoplasmic ribosomes
 - Asexual mode of reproduction
 - Cell wall made up of peptidoglycan
 - Chemoautotrophic mode of nutrition
103. Which of the following is a feature of archaeobacteria that makes them different from other prokaryotes?
- Presence of amino acid with sugar in cell wall
 - Presence of lipids in cell membrane
 - Presence of introns in the genetic sequence
 - Synthesis of organic material from inorganic substances with the help of chemical energy
104. Read the following Assertion (A) and Reason (R) statements.
Assertion (A) : Mode of nutrition in *Euglena* is said to be mixotrophic.
Reason (R) : Nutrition in *Euglena* is photoautotrophic. However in absence of light, it is capable of getting nutrition from dead and decaying matter.
In the light of the above statements, select the **correct** option.
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - Both (A) and (R) are false
105. Stiff cellulosic plates on the outer surface of cell wall are found in
- Euglena*
 - Dinoflagellates
 - Mycoplasma
 - Archaeobacteria

106. Which of the following organelles is associated with autophagy?

- (1) Endoplasmic reticulum
- (2) Golgi body
- (3) Lysosome
- (4) Mitochondrion

107. Which of the following features is **not** common between mitochondria and chloroplasts?

- (1) Double membrane bound
- (2) ATP synthesis
- (3) Semi-autonomous nature
- (4) Synthesis of carbohydrates

108. Match the Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
a.	Luminal compartment	(i)	Modified Golgi apparatus
b.	SER	(ii)	Endoplasmic reticulum
c.	Acrosome	(iii)	Muscle contraction by uptake and release of Ca^{2+}
d.	Residual bodies	(iv)	Formed by lysosomes

- (1) a(i), b(iii), c(ii), d(iv)
- (2) a(ii), b(iii), c(i), d(iv)
- (3) a(ii), b(iv), c(i), d(iii)
- (4) a(iii), b(i), c(ii), d(iv)

109. Which of the following group of plants is called "Amphibians of the plant kingdom"?

- (1) Gymnosperms
- (2) Bryophytes
- (3) Pteridophytes
- (4) Algae

110. Mosses differ from liverworts as the former

- (1) Have spores, which germinate to form free living gametophytes
- (2) Possess thalloid plant body
- (3) Have multicellular and branched rhizoids
- (4) Undergo asexual reproduction by means of fragmentation

111. Which of the following is a heterosporous pteridophyte?

- (1) *Dryopteris*
- (2) *Equisetum*
- (3) *Salvinia*
- (4) *Lycopodium*

112. Select the **mismatched** pair.

- (1) Amoeboid protozoan - *Entamoeba*
- (2) Flagellated protozoan - *Trypanosoma*
- (3) Ciliated protozoan - *Amoeba*
- (4) Sporozoan - *Plasmodium*

113. All of the following fungi produce non-motile asexual spores and have septate mycelium, **except**

- (1) *Penicillium*
- (2) *Collectotrichum*
- (3) *Neurospora*
- (4) *Albugo*

114. Prothallus of pteridophytes is

- (a) Unicellular
 - (b) Inconspicuous
 - (c) Photosynthetic
 - (d) Thalloid
- The **correct** ones are

- (1) a, b and d
- (2) b, c and d
- (3) a, c and d
- (4) a, b and c

115. Which of the following features are seen in both the first embryophytes and the seedless vascular plants?

- a. Thalloid main plant body
- b. Need of water for fertilization
- c. Prevent soil erosion
- d. Presence of rhizoids
- e. Homospory
- f. Flagellated antherozoid

Choose the **correct** answer from the following options.

- (1) a, b and d only
- (2) b, c and e only
- (3) c, d and e only
- (4) b, c, e and f

116. Starch-containing colourless plastids are

- (1) Chromoplasts
- (2) Aleuroplasts
- (3) Elaioplasts
- (4) Amyloplasts

117. Find the **odd** one w.r.t. the function of the microtubules.

- (1) They help in spindle and aster ray formation
- (2) They help in intracellular transport of nutrients
- (3) They are involved in pseudopodia formation
- (4) The position of the future cell plate is determined by them

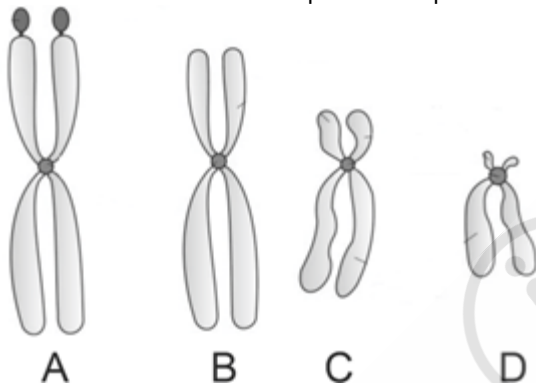
118. Under his own designed microscope, living cells were first observed by

- (1) Robert Hooke
- (2) Anton von Leeuwenhoek
- (3) Robert Brown
- (4) Palade

119. The main constituent of the fungal cell wall is

- (1) Cellulose
- (2) Peptidoglycan
- (3) Chitin
- (4) Mannans

120. In the given diagram of chromosomes. Identify the ones in which their arms are referred to as p-arm and q-arm



- (1) A and B
- (2) B and C only
- (3) B, C and D
- (4) C and D only

121. The small fragment of chromosome that appears due to non-staining secondary constriction is called

- (1) Centromere
- (2) Satellite
- (3) Kinetochore
- (4) Telomere

122. Identify the **correct** ones w.r.t. centriole.

- a. It has 9 + 2 arrangement of microtubules.
- b. There are 9 peripheral triplets.
- c. The centre of the centriole possess a rod shaped mass of tubulin known as massule.
- d. It has a cartwheel appearance.
- e. It is surrounded by pericentriolar satellite.

- (1) a, b and d
- (2) b, d and e
- (3) b, c and d
- (4) a, c and e

123. Na^+/K^+ pump in the plasma membrane is associated with

- (1) Osmosis
- (2) Active transport
- (3) Passive transport
- (4) Simple diffusion

124. Which of the following is considered to be polymorphic cell organelle?

- (1) Lysosome
- (2) Chloroplast
- (3) Golgi body
- (4) Endoplasmic Reticulum

125. Read the following statements and choose the **correct** option.

Statement A : Chromatin is named so because of its ability to get stained with certain acidic dyes.

Statement B : Chromatin is composed of DNA and histones only.

- (1) Only statement A is incorrect
- (2) Only statement B is incorrect
- (3) Both the statements are correct
- (4) Both the statements are incorrect

126. Spores of mosses, on liberation, germinate to produce

- (1) Protonema
- (2) Sporophyll
- (3) Sporophyte
- (4) Prothallus

127. *Selaginella* and *Equisetum* belong to the class,

- (1) Psilopsida
- (2) Lycopsida and Sphenopsida, respectively
- (3) Sphenopsida and Pteropsida, respectively
- (4) Pteropsida

128. Find the **incorrect** option w.r.t. *Sphagnum*.

- (1) *Sphagnum* fills ponds and lakes with soil
- (2) Dry *Sphagnum* has great water absorbing capacity
- (3) Sexual reproduction in *Sphagnum* occurs by means of gemmae
- (4) *Sphagnum* is responsible for the formation of peat

129. Read the following statements and choose the **correct** option.

Statement A : In gymnosperms, leaves can be simple or compound.

Statement B : In gymnosperms, male and female gametophytes have a free living existence.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect

130. Identify the following statements as True (T) or False (F) and choose the **correct** option.

A. Gymnosperms generally have tap roots.

B. In conifers, leaves are needle-like, having reduced surface area, thin cuticle and sunken stomata.

C. Gymnosperms are heterosporous.

D. In gymnosperms, pollination is carried out by air currents.

	A	B	C	D
(1)	F	T	F	T
(2)	T	F	T	T
(3)	T	T	F	F
(4)	F	F	T	T

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

131. Which of the following scientists demonstrated that the extract of the infected plant of tobacco could cause infection in healthy plants?

- (1) D. J. Ivanowsky
- (2) M. W. Beijerinck
- (3) W. M. Stanley
- (4) T.O. Diener

132. Which of the following plants is an example of a parasitic plant?

- (1) Bladderwort
- (2) Venus fly trap
- (3) *Cuscuta*
- (4) Sugarcane

133. Find the **incorrect** match.

- (1) Zoospore – Members of Phycomycetes
- (2) Endogenously produced meiospores – *Claviceps purpurea*
- (3) Secondary dikaryotic mycelium – *Alternaria solani*
- (4) Basidiocarps - *Ustilago*

134. Find the **incorrect** statement w.r.t. sub-viral agents.

- (1) Viroids are infectious DNA particles, devoid of a protein coat
- (2) Viroids are obligate parasites
- (3) Viroids are known to cause diseases in plants
- (4) Prions are similar in size to viruses

135. Which of the following is a unicellular organism having 80S ribosomes and cell organelles?

- (1) *E. coli*
- (2) *Amoeba proteus*
- (3) *Nostoc*
- (4) *Mycoplasma*

ZOOLOGY

136. Select the correct statement.

- (1) All cyclostomes have larvae which are exclusively marine.
- (2) All amphibians are limbless and exhibit external fertilisation.
- (3) All reptiles are exclusively terrestrial animals.
- (4) All fishes have a streamlined body.

137. Organ-level of body organisation is found in the members belonging to the phylum

- (1) Porifera
- (2) Coelenterata
- (3) Ctenophora
- (4) Platyhelminthes

138. Hookworm belongs to the phylum

- (1) Aschelminthes
- (2) Annelida
- (3) Platyhelminthes
- (4) Ctenophora

139. Which of the following is **incorrect** w.r.t. *Nereis*?

- (1) Belongs to the similar phylum as *Pheretima*
- (2) Aquatic habitat and presence of metamerites
- (3) Exhibit parapodia for locomotion
- (4) Monoecious in nature

140. All of the following belong to the phylum Echinodermata, except

- (1) Sea urchin
- (2) Sea lily
- (3) Sea cucumber
- (4) Sea hare

141. Choose the correct statement.

- (1) Lipids and monosaccharides are generally water-soluble.
- (2) Trihydroxy propane is a conjugated lipid.
- (3) A triglyceride is composed of one glycerol molecule and two fatty acids.
- (4) The neural tissues have lipids with more complex structures.

142. Complete the analogy by selecting the correct option.

Direct development : *Labeo* :: Indirect development : _____.

- (1) *Hyla*
- (2) *Hemidactylus*
- (3) *Aptenodytes*
- (4) *Balaenoptera*

143. Which of the following is a marine animal that possesses placoid scales but is devoid of operculum and air bladder?

- (1) *Scoliodon*
- (2) *Exocoetus*
- (3) *Hippocampus*
- (4) *Clarias*

144. Choose the animal that contains notochord only in its larval tail.

- (1) *Ascidia*
- (2) *Branchiostoma*
- (3) *Petromyzon*
- (4) *Myxine*

145. The marine cartilaginous fish which has electric organs is

- (1) *Trygon*
- (2) *Torpedo*
- (3) *Pristis*
- (4) *Carcharodon*

146. The type of epithelium found in the inner lining of ducts of salivary glands is also present in the

- (1) Inner lining of pancreatic ducts
- (2) Lining of fallopian tubes
- (3) Walls of alveoli
- (4) Inner wall of small intestine

147. Select the incorrect statement.

- (1) Areolar tissue is present beneath the skin.
- (2) Adipose tissue contains fibroblasts and adipocytes.
- (3) Dense irregular connective tissue contains collagen fibres.
- (4) Loose connective tissue has a compact and dense matrix.

148. Neuroglial cells

- (1) Can transmit impulses
- (2) Are structural and functional units of neural tissue
- (3) Make less than one half the total volume of the neural tissue
- (4) Protect and support neurons

149. Consider the following features

- (a) Join different bones together in the region of joints.
 - (b) Contain collagen fibres that are present in rows between many parallel bundles of fibres
 - (c) Attach skeletal muscle to bone
 - (d) Categorized as dense irregular connective tissue
- Select the option which represents only the correct features w.r.t. ligaments.

- (1) Only (a) and (b)
- (2) Only (a) and (c)
- (3) Only (b) and (c)
- (4) Only (a), (b) and (d)

150. Match column I with column II and choose the correct option.

	Column I		Column II
a.	Tight junctions	(i)	Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells
b.	Adhering junctions	(ii)	Present in cardiac muscle tissue
c.	Gap junctions	(iii)	Perform cementing to keep neighbouring cells together
d.	Communication junctions	(iv)	Help to stop substances from leaking across a tissue

- (1) a(iii); b(iv); c(ii); d(i)
 (2) a(iv); b(iii); c(i); d(ii)
 (3) a(iii); b(iv); c(i); d(ii)
 (4) a(iii); b(ii); c(iv); d(i)

151. Which among the following is common for chiton, *Sepia* and pearl oyster?

- (1) Segmented body
 (2) Presence of soft and spongy layer over visceral hump
 (3) Monoecious nature
 (4) Viviparity and direct development

152. All of the following features are associated with ctenophores, except

- (1) Exclusively marine habitat
 (2) Bioluminescence
 (3) External fertilisation and direct development
 (4) Extracellular and intracellular digestion

153. Which among the following constitutes negligible percentage weight in human body as compared to the Earth's crust?

- (1) Silicon
 (2) Sodium
 (3) Nitrogen
 (4) Oxygen

154. Select the incorrect one w.r.t. proteins

- (1) Heteropolymeric nature
 (2) Can act as biological catalysts
 (3) Can act as cellular receptors
 (4) Obtained in acid-soluble pool

155. In the B-DNA model, the two antiparallel polynucleotide strands are joined together by

- (1) Glycosidic bonds
 (2) Phosphodiester bonds
 (3) Sugar-phosphate bonds
 (4) Hydrogen bonds

156. Comprehend the following

- a. It is a structural polysaccharide
 b. Found in exoskeleton of arthropods
 c. It is the polymer of glucose
 Choose the **correct** option w.r.t. chitin

- (1) b and c only
 (2) a and b only
 (3) a, b and c
 (4) a and c only

157. Read the statements given below and choose the correct option.

Statement(A): Adult human haemoglobin is a tetramer formed of more than 2 subunits.

Statement(B): Proline and valine are sulphur containing amino acids.

- (1) Both statements (A) and (B) are correct
 (2) Both statements (A) and (B) are incorrect
 (3) Only statement (A) is correct
 (4) Only statement (B) is correct

158. Complete the analogy by selecting the correct option.

Acidic amino acid : Glutamic acid :: Basic amino acid : _____.

- (1) Alanine
 (2) Aspartate
 (3) Lysine
 (4) Glycine

159. Enzymes catalysing the transfer of a group (other than hydrogen) between a pair of substrates, belong to the class _____ of enzymes.

Select the correct option to fill in the blank.

- (1) I
 (2) II
 (3) III
 (4) IV

160.Assertion(A): If you add iodine solution to a sample containing starch, it gives blue colour.

Reason(R): Starch forms helical secondary structures in which it can hold I_2 molecules.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true, (R) is false
- (4) (A) is false, (R) is true

161.Choose the correct match.

(1)	Glucose	–	Hexose sugar that contains a free carboxyl group at the 2C
(2)	Cholesterol	–	Contains 5 homocyclic rings
(3)	Lecithin	–	Polymeric substance
(4)	Cytidine	–	Nitrogenous base + Pentose sugar

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

162.How many of the molecules mentioned in box below will be found in the acid-soluble pool obtained upon chemical analysis of a living tissue?

Collagen, Cellulose, Glycogen, Fructose, Adenine

Select the correct option.

- (1) Five
- (2) Two
- (3) Three
- (4) Four

163.Match column I with column II w.r.t. secondary metabolites and select the correct option.

	Column I		Column II
a.	Drugs	(i)	Lemon grass oil
b.	Lectins	(ii)	Morphine, Codeine
c.	Alkaloids	(iii)	Vinblastin, Curcumin
d.	Essential oil	(iv)	Concanavalin A

- (1) a(iii) ; b(iv) ; c(i) ; d(ii)
- (2) a(iii) ; b(iv) ; c(ii) ; d(i)
- (3) a(iv) ; b(iii) ; c(i) ; d(ii)
- (4) a(i) ; b(ii) ; c(iii) ; d(iv)

164.Read the statements given below:

- (a) The activity of succinic dehydrogenase declines in the presence of malonate.
 - (b) The non-protein part of an enzyme, that joins with a co-enzyme is called the apoenzyme.
 - (c) Low temperature preserves the enzyme in its temporarily inactive state
 - (d) Enzymes remain unchanged after completion of a chemical reaction and can be re-utilised.
- Choose the correct option.

- (1) Only (a) and (b) are correct
- (2) Only (a), (b) and (c) are incorrect
- (3) Only (c) and (d) are incorrect
- (4) Only (a), (c) and (d) are correct

165.Choose the correct statement w.r.t. frogs

- (1) External ear is represented by the auricular muscles
- (2) A muscular, bilobed tongue is used to capture the prey
- (3) On land, cutaneous respiration is exclusively used to supply oxygen to the tissues
- (4) Unlike humans they lack the common bile duct.

166.Read the following statements and choose the correct option.

Statement A : In some countries, the muscular legs of frogs are used as food by man

Statement B : The system for control and coordination is highly evolved in frogs

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

167.Male and female cockroaches can be distinguished by the

- (1) Presence of anal cerci in former
- (2) Absence of caudal style in latter
- (3) Absence of uricose glands in former
- (4) Presence of titillator in latter

168.In frogs, the hind brain consists of

- (1) Pons and cerebellum
- (2) Pons, cerebellum and medulla oblongata
- (3) Cerebellum and medulla oblongata
- (4) Pairs of optic lobes, cerebellum and medulla oblongata

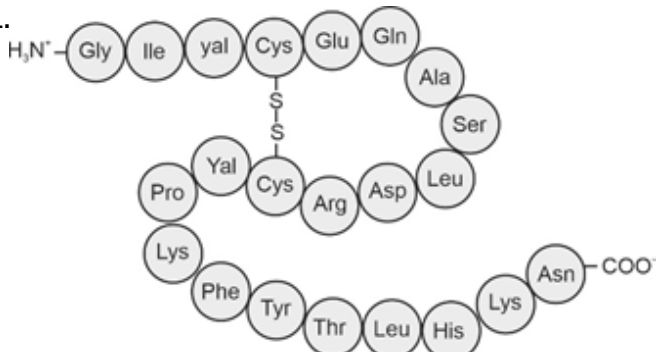
169.In cockroach, with the help of several ommatidia, it can receive several images of an object. This kind of vision is

- (1) Mosaic vision with more sensitivity and less resolution
- (2) Mosaic vision with less sensitivity and high resolution
- (3) Nocturnal vision with more sensitivity and high resolution
- (4) Nocturnal vision with less sensitivity and less resolution

170. Male frogs can be distinguished from the female frogs by which of the following characters?

- (1) Presence of vocal cords in males
- (2) Presence of copulatory pad on the first digit of hind limb of males
- (3) Presence of sound producing vocal sacs in males
- (4) Five webbed digits in the hind limbs of females

171.



Consider the structure of a protein given above and choose the incorrect option w.r.t it.

- (1) Glycine is the N-terminal amino acid
- (2) There is no aromatic amino acid in the chain
- (3) The 'R' group in the 8th amino acid is hydroxy methyl
- (4) Complete hydrolysis of the given structure will yield more than 20 amino acid molecules.

172. Assertion (A): Platyhelminths are mostly endoparasites, found in animals including human beings.

Reason (R): Hooks and suckers are present in parasitic flatworms.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true, (R) is false
- (4) Both (A) and (R) are false

173. Choose the incorrect match

	Organism	Character/structure	Phylum
(1)	<i>Taenia</i>	True metamerism	Platyhelminthes
(2)	<i>Ascaris</i>	Muscular pharynx	Aschelminthes
(3)	<i>Hirudinaria</i>	Nephridia	Annelida
(4)	<i>Locusta</i>	Jointed appendages	Arthropoda

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

174. In the members of which of the following phyla, excretory system is absent?

- (1) Echinodermata
- (2) Hemichordata
- (3) Mollusca
- (4) Annelida

175. Read the statements given below:

- (a) Cyclostomes have a cartilaginous vertebral column.
- (b) In the members of class Chondrichthyes, the notochord is persistent throughout their life
- (c) All chordates have a closed circulatory system with a dorsal heart.

Select the option that represents only the correct statements.

- (1) (a), (b) and (c)
- (2) (a) and (b) only
- (3) (a) and (c) only
- (4) (b) and (c) only

176. Match column I with column II w.r.t. cockroaches and select the correct option.

	Column I		Column II
a.	Testes	(i)	6 th -7 th abdominal segments of males
b.	Ovary	(ii)	6 th abdominal segment of females
c.	Mushroom-shaped gland	(iii)	2 nd -6 th abdominal segments of females
d.	A pair of spermathecae	(iv)	4 th -6 th abdominal segments of males

- (1) a(iv); b(iii); c(ii); d(i)
- (2) a(iv); b(iii); c(i); d(ii)
- (3) a(iii); b(iv); c(ii); d(i)
- (4) a(iii); b(iv); c(i); d(ii)

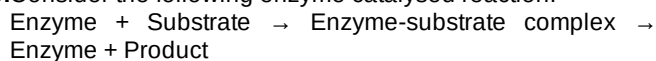
177. Name the special venous connection which is present in both humans and frogs and carries deoxygenated blood from GIT to the liver.

- (1) Hepatic portal system
- (2) Renal portal system
- (3) Coronary circulation
- (4) Hypophyseal portal system

178. Select the incorrect statement w.r.t. *Rana tigrina*

- (1) Adult frog is a ureotelic animal
- (2) Each kidney is composed of several structural and functional units called uriniferous tubules
- (3) Thin walled urinary-bladder present dorsal to rectum receives opening of ureters
- (4) Kidneys are red and bean like structures situated posteriorly on both sides of vertebral column

179. Consider the following enzyme catalysed reaction:



Amongst all components mentioned above, which one has the highest energy level?

- (1) Substrate
- (2) Product
- (3) Enzyme
- (4) Enzyme-substrate complex

180. Amongst all, select the nucleoside which is exclusively present in RNA but absent in DNA.

- (1) Adenosine
- (2) Guanosine
- (3) Thymidine
- (4) Uridine

